

Model Proposal

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The overall topic behind our two models is the relationships between car crashes in the Charlotte Region and the population change in the city over the years. Our regression model idea was to predict what percentage of crashes increases (exponential, linear) based on how many people move to Charlotte, using population change by year and yearly crash reports for predictions. The model will take the two datasets and merge them together based on the same key variable year allowing us to align population trends directly with crash statistics for the same time periods. This merged dataset will serve as the foundation for training and evaluating our regression model. The target variable is the number of traffic accidents and is based on the following features: population change, population growth percentage, and crash growth percentage.

Our proposed classification model will be developing a binary classification model to predict high versus low traffic fatality risk years in Mecklenburg County, North Carolina. Using historical transportation data spanning 1970–2024, the model will use key road safety indicators. These include alcohol-related accidents, speed violations, motorcyclist injuries, and pedestrian accidents, alongside annual population estimates to classify each year as either high or low fatality risk relative to a median threshold. We will incorporate per-capita normalization through county population estimates so the model accounts for the influence of population growth on raw accident counts. This will allow for a more accurate safety risk estimate. The insights generated could potentially inform data-driven traffic safety policy, help guide resource allocation, and serve as a replicable framework for other North Carolina counties as additional transportation data becomes available.

Datasets Used:

[Crash data set](#)

[Pop dataset](#)